

Worksheet 1

For Practice in Introductory Calculus (NOT MANDATORY)

October 16, 2025

Remark: In formal exams, when applying L'Hôpital's Rule, it is standard practice to indicate its use by placing "L'H" or "L'Hôp" above the equals sign. For example:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} \stackrel{L'H}{=} \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1$$

This worksheet does not always include this notation for brevity, but you should use it in your exams.

Note: Problems marked with an asterisk (*) are optional and are less likely to be tested in the exams. You can safely skip them if you don't want to spend much time on the worksheet. Have fun and good luck!—LiZi

1. Inverse Functions

- *1. Given the function $f(x) = x^3 + 6x^2 + 12x + 5$, find its inverse function $f^{-1}(x)$.
- *2. Given the function $f(x) = \frac{x}{\sqrt{1-x^2}}$ with a domain of $(-1, 1)$, find its inverse function $f^{-1}(x)$.

- *3. If $f(x) = \sqrt[3]{x + \sqrt{x^2 + 1}} + \sqrt[3]{x - \sqrt{x^2 + 1}}$, find $f^{-1}(x)$.

$$\text{Hint: } a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

- *4. Consider the function $f(x) = x^3 - 3x$.

- (a) By restricting the domain to $x \in [1, 2]$, the function is strictly increasing. Find the inverse function $f^{-1}(x)$ for this restricted domain.

$$\text{Hint: } \cos(3\theta) = 4\cos^3\theta - 3\cos\theta.$$

- (b) Now, consider the same function but restrict its domain to $x \in [2, \infty)$, where it is also strictly increasing. Find the inverse function $f^{-1}(x)$ for this new restricted domain.

Hint: The trigonometric substitution from part (a) is not applicable here due to the new domain. However, a similar identity exists for the hyperbolic cosine function, defined as $\cosh(t) = \frac{e^t + e^{-t}}{2}$. The identity is $\cosh(3t) = 4\cosh^3 t - 3\cosh t$.

2. Limits

5. $\lim_{x \rightarrow \infty} \left(\sqrt{x + 3\sqrt{x}} - \sqrt{x - \sqrt{x}} \right)$

6. $\lim_{x \rightarrow \infty} \frac{x^2 + 5}{2x + 1} \sin \left(\frac{3}{x} \right)$

7. $\lim_{x \rightarrow 0} x^{-3} \left[\left(\frac{2 + \cos x}{3} \right)^x - 1 \right]$

8. $\lim_{x \rightarrow \infty} (\ln x)^{\frac{1}{x}}$

9. $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$

10. $\lim_{x \rightarrow 0^+} \left[\frac{(1+x)^{\frac{1}{x}}}{e} \right]^{\frac{1}{x}}$

11. $\lim_{x \rightarrow \alpha} \frac{\ln x - \ln \alpha}{x - \alpha}, \quad (\alpha > 0)$

12. $\lim_{x \rightarrow 0^+} \frac{1 - \sqrt{\cos x}}{x(1 - \cos \sqrt{x})}$

13. $\lim_{n \rightarrow \infty} n(\sqrt[n]{x} - 1), \quad (x > 0)$

14. Prove that $\lim_{n \rightarrow \infty} \sum_{i=0}^n \frac{n}{n^2 + i\pi} = 1$.

15. Prove that $\lim_{n \rightarrow \infty} \sum_{i=0}^n \ln \left(1 + \frac{i}{n^2} \right) = \frac{1}{2}$.

Hint: $x > \ln(1+x) > x - \frac{x^2}{2}$

16. Calculate the limit:

$$\lim_{x \rightarrow 0} \frac{e^{x^2} - \cos(x) - \frac{3}{2}x^2}{x^4}$$

17. Let $a, b > 0$. Calculate the limit:

$$\lim_{x \rightarrow 0} \left(\frac{a^x + b^x}{2} \right)^{\frac{1}{x}}$$

18. Evaluate the following limit related to Stirling's approximation:

$$\lim_{n \rightarrow \infty} \frac{\sqrt[n]{n!}}{n}$$

Hint: You may use Stirling's approximation for the factorial, which states that for large n , $n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$.

(For students familiar with integration, consider using a Riemann sum to approximate $\ln(n!)$. This provides an alternative approach and reflects the core idea behind the historical development of Stirling's approximation.)

- *19. Calculate the following second-order limit involving the number e :

$$\lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e + \frac{e}{2}x}{x^2}$$

The intended solution for this problem requires Taylor Expansions. If you haven't covered this topic yet, feel free to skip to the next question. For those looking for a challenge, you can attempt a solution using L'Hôpital's Rule, but be warned—the calculation is very long and annoying!

20. Evaluate the following limits:

$$\lim_{x \rightarrow 0^+} [\cos(2x)]^{1/x^2}$$

21. Evaluate the following limits:

$$\lim_{t \rightarrow \infty} \left[t \ln \left(1 + \frac{3}{t} \right) \right]$$

3. Derivatives and Continuity

22. The Folium of Descartes is given by the equation $x^3 + y^3 = 6xy$. Find the value of the second derivative, $\frac{d^2y}{dx^2}$, at the point $(3, 3)$.
23. The Witch of Agnesi curve is given by the equation $y = \frac{8}{x^2+4}$. Find the coordinates of its inflection points.

24. A cycloid is given by the parametric equations:

$$x = r(\theta - \sin \theta), \quad y = r(1 - \cos \theta)$$

Find the second derivative, $\frac{d^2y}{dx^2}$, as a function of θ .

25. Given $\cos(x^2 + 2y) + xe^{y^2} = 1$, find $\frac{dy}{dx}$ by implicit differentiation.

26. Given $f : \mathbb{R} \rightarrow \mathbb{R}$, with

$$f(x) = \begin{cases} x^4 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

(a) Calculate $f'(x)$.

(b) Calculate $f''(0)$.

27. Suppose that $f(x)$ is continuous near $x = 0$, satisfying $f(x) - \frac{1}{2}f(\frac{x}{2}) = x^2$. What is $f(x)$?

28. Suppose

$$f(x) = \begin{cases} \frac{x}{1+e^{\frac{1}{x}}} & (x \neq 0) \\ 0 & (x = 0) \end{cases}$$

Prove that $f(x)$ is continuous on $\mathbb{R}$.

29. Suppose that f satisfies the equation

$$f(x+y) = f(x) + f(y) + x^2y + xy^2 \quad \text{for all } x, y \in \mathbb{R}$$

Suppose further that

$$\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$$

(a) Calculate $f'(0)$.

(b) Calculate $f'(x)$.

4. Applications of Derivatives & Theorems

30. Suppose we know that $f(x)$ is continuous and differentiable on the interval $[-6, -1]$, that $f(-6) = -23$ and that $f'(x) \geq -4$. What is the smallest possible value for $f(-1)$?